Unit 2

Regular Expressions:

- method to represent a language
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- It is similar to Boolean expressions logical expressions, Algebraic expressions
- In TOC we use Regular expressions to represent a language.
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Regular Expression is related with Regular languages.

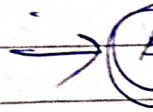
What is Regular Languages?

- Regular Language:
  All the languages accepted by finite Automata is called Regular Languages.
  Ex: Finite Automata:
  DFA, NFA, ε-NFA, etc.
  ⇒ Different kinds of machines: DFA, NFA
- Regular Language represented by Regular Expression
  - which is also used in compiler Design.
  - $L = \{\varepsilon, a, aa, aaa\}$

$L = a^*$ Infinite language

put * as 0, 1, 2, 3, ...

Finite automata ⇒



Corresponding to Finite Automata or corresponding with Regular languages we represent Regular Expressions

Definition of Regular Expression :

Let 'R' be an Regular Expression over alphabet Σ if R is : (a,b)

1) ϵ or ϕ is Regular Expression denoting set $\{\epsilon\}$ ⇒ collection of NULL string.

$R = \epsilon \quad L(R) = \{\epsilon\}$

2) ϕ, is Regular Expression the empty set $\{\}$

$R = \phi \quad L(R) = \{\}$

zero NO strings | NO strings without any symbol.

(3) For each symbol $a \in \Sigma$, a is regular expression. denoting set $\{a\}$.

$R = a \quad L(R) = \{a\}$

(4) Primitive – a minimum Language created by Automata.

$L(R) = \{\epsilon\}$
$L(R) = \{\}$
$L(R) = \{a\}$
} primitive Language

(4) Union of RE is also Regular,
Union : $R_1 \cup R_2$ = Regular ex: $a \cup b = \{a, b\}$
(5) Concatenation : $R_1 R_2$ = Regular
(6) Kleene closure of RE = Regular,
$a^* = \{\}$ * : we put 0, 1, 2, 3 in place of *.
$(RR)^*$ = regular

When we write Regular expression of finite languages we must do union.

$L = \{\epsilon, a, aa, aaa, aaaa, \ldots\}$

(7) If R is regular (R) is also Regular.

ex: $(a+b)^* = \{\epsilon, a, b, ab, ba, aa, bb, abab\}$.

RE = Language

RE = Finite Automata

Competitive Exam:

Regular Expression for finite language

finite Languages are always regular

1) Regular ✓
2) Finite Automata ✓ possible
3) Regular Expression ✓ possible.

Finite Languages:

To write Regular Expression of finite Automata

1) Language with No string. $\{\}$ ϕ — R.E ϕ
2) Language with length 0 $\{\epsilon\}$ ϵ, Λ
3) language with length 1 $\{a, b\}$ $(a+b)$

$a \cup b$ = Union operation = either a or b.

4) Language with length 2 $\{aa, bb, ab, ba\}$
$\{aa+ab+ba+bb\}$
$\{a(a+b), b(a+b)\}$
$\{(a+b)(a+b)\}$
5) language with length 3 $(a+b)(a+b)(a+b) = \{aaa, aab, aba, abb, baa, bbb\}$
6) Atmost 1, 0 or 1 $\{\epsilon, a, b\}$ $\{\epsilon + a + b\}$
$\{\epsilon, a, b\}$ $\{\epsilon + a + b)(\epsilon + a + b)$

b | a | a | b | a | ...

$(a+b)^*$ = All strings are possible
= universal string
= {ε, a, b, aa, ab ...}

7) No. more than 2 b's and 1 a...
{ε, a, b, ab, ba, abb, bab, bba}

R.E = (ε + a + b + ab + ba + abb + bab + bba)

~~(ε + a + b) ab (ε + b + b)~~

simplify it ~

Regular Expression for Infinite language

Regular expression of Infinite languages are represented by '*', '+' [pwq]

may be regular or maynot regular

ex: a^* ⇒ pwr 0, 1, 2, 3, ... in place of *

$a^0 = \epsilon$, $a^1 = a$, $a^2 = aa = (\epsilon, aaa)$

ex $a^+ = aa^*$, pwr 0, 1, 2, 3 (a, aa, ...)
Not included 'ε' in this R.E.

$\Sigma (a, b)$

R.E

1) All strings having single 'b' a^*ba^*
2) All strings having atleast 1 b $(a+b)^*b(a+b)^*$
3) All strings having "bbbb" as substring $(a+b)^*bbbb(a+b)^*$
4) All strings end with ab $(a+b)^*ab$
5) starting with ba. $ba(a+b)^*$
6) starting with a & ending with a. $a(a+b)^*a$
7) starting a $(a+b)^*a(a+b)^*$
8) starting with a & end with b $(a(a+b)^*b$
9) ~~b~~ starting with b & end with a $b(a+b)^*a)$

10) exactly two b's $a^*ba^*ba^*$